University of Warsaw

Faculty of Economic Sciences

Jan Muchorowski

Student ID: 440776

Modeling fiscal-monetary regime switching in Poland

Bachelor's Thesis

in Economics

Supervisor:

Ph.D. Piotr Żoch

Department of Statistics and Econometrics

Faculty of Economic Sciences

University of Warsaw

Warsaw, 2026

Oświadczenia

Oświadczenie kierującego pracą

Oświadczam, że niniejsza praca została przygotowana pod moim kierunkiem i stwierdzam, że spełnia ona warunki do przedstawienia jej w postępowaniu o nadanie tytułu zawodowego.

Data	Podpis kierującego pracą
______________________	______________________

Oświadczenie autora pracy

Świadom odpowiedzialności prawnej oświadczam, że niniejsza praca dyplomowa została napisana przeze mnie samodzielnie i nie zawiera treści uzyskanych w sposób niezgodny z obowiązującymi przepisami.

Oświadczam również, że przedstawiona praca nie była wcześniej przedmiotem procedur związanych z uzyskaniem tytułu zawodowego w wyższej uczelni.

Oświadczam ponadto, że niniejsza wersja pracy jest identyczna z załączoną wersją elektroniczną.

Data	Podpis autora pracy
______________________	______________________

Oświadczenie o wykorzystaniu w pracy systemów sztucznej inteligencji (SI)

Podczas przygotowania tej pracy autor użył modelu językowego Claude w celu poprawienia części kodu oraz poprawienia błędów składniowych i językowych.

Autor pracy dokonał weryfikacji treści i redakcji tekstu pracy zgodnie z jej założeniami, prezentując własny wkład intelektualny i biorąc pełną odpowiedzialność za ostateczną postać pracy.

Podpis Autora / Autorów

Abstract:

This thesis investigates the existence of monetary- and fiscally-led policy regimes in Poland between 2000 and 2025 using Markov-switching macroeconomic models. First, I estimate the Markov-Switching VAR model to identify possible regime changes in inflation, output growth, interest rates, debt, and the primary balance. Second, I develop the Markov-Switching DSGE model with rational expectations.

The results suggest the presence of two distinct regimes. The first is characterized by stronger monetary responses to inflation and greater fiscal discipline, while the second exhibits weaker monetary reactions and more persistent fiscal imbalances. The estimated probabilities indicate that fiscally-led dynamics became more likely during periods of economic stress, particularly in 2019–2022. Impulse response analysis further shows that the transmission of monetary and fiscal shocks differs substantially across regimes.

Keywords:

fiscal dominance, monetary dominance, Markov-switching VAR, Markov-switching DSGE

Erasmus Classification:

Economics (14300)

JEL Classification:

E31, E52, E62, H63

Title in Polish:

Modelowanie zmian reżimów fiskalno-monetarnych w Polsce.

Contents

ARTYKUŁ NAUKOWY

”Sustained high inflation is always and everywhere a fiscal phenomenon, in which the central bank is a monetary accomplice”

– Tom Sargent

1 Introduction

The recent inflationary episode—the most significant in the past four decades—has attracted substantial attention in the economic literature. As Reis (2026) argues, several conventional explanations have been proposed. Following the end of pandemic-related restrictions, workers returned to the labor market, while governments provided additional support to household incomes through extensive fiscal transfers. At the same time, global supply chains remained severely disrupted due to the breakdown of international trade routes. The interaction of these demand- and supply-side shocks generated strong inflationary pressures, which were further intensified after Russia's full-scale invasion of Ukraine in 2022, as emphasized by Powell (2025) in his Jackson Hole speech.

The role of fiscal policy—and the reluctance of monetary authorities to raise interest rates in the aftermath of the combined demand and supply shocks, partly out of concern that tighter monetary conditions could undermine post-Covid relief and recovery programs—has also become the subject of extensive debate (Bianchi & Barro 2026). The issue of rising public debt has long occupied a central place in macroeconomic discussions; however, recent years have renewed interest in the concepts of fiscal and monetary dominance, the possibility of regime changes between them, and the macroeconomic consequences associated with such shifts.

This paper builds on the assumption, shared by many economists (Leeper 1991, Sims 1994, Woodford 1994), that government debt and its management play a central role in shaping real economic outcomes. This line of research has resulted in characterization of two possible ways of reacting to inflation and debt: the policy may be monetary–led (when the central bank chooses interest rates and the government adjusts its spending to keep debt in check) or fiscally–led (when the government chooses the path of spending and central bank keeps the interest rate below inflation, thus keeping the real debt in check).

The existence of fiscal and monetary regimes, together with their historical decomposition, has already been documented for the United States (Bianchi & Melosi 2017) and the United Kingdom (Webb 2025). However, no comparable analysis has yet been conducted for Poland. Although there has been studies examining regime switching within Taylor-rule specifications (Baranowski & Sztaudynger 2016), none has combined monetary–fiscal interactions with the possibility of multiple parametrs ruling both fiscal and monetary policy in a DSGE framework. Consequently, this paper constitutes the first attempt to identify and quantify the existence of such regimes in the Polish economy.

My main objective in this study is to determine whether regime changes occurred in Poland, to quantify the probability of their existence, and to evaluate how different regimes affected key macroeconomic variables, namely interest rates, inflation, real GDP, and debt. In the next section I review the existing literature on policy regimes, their classification, and quantitative analyses of their macroeconomic implications. Subsequently, I develop two models. First, an MS-VAR model is estimated in order to obtain priors for the Markov chain governing regime transitions and to examine whether the Polish economy exhibits regime-switching behavior. Second, I introduce the main empirical framework of the paper, an MS-DSGE model. In the subsequent section I present the results, including calibrated parameters, impulse response functions across regimes, which helps to explain the main differences between models. In the final section I offer conclusion and additional thoughts.

In this paper I show that according to the developed model there was a high probability of Poland entered into fiscally-led regime twice from 2000: first around the year 2006, after entering into European Union, and second from 2019 to 2023, especially after Covid-19 pandemic and later during inflation crisis. During those episodes, the reaction of central bank to inflation was weaker, while the government did not raise primary surpluses to manage its debt. This result serves to raise awareness of the changing policy regimes, their results and the importance of fiscal–monetary interactions.

2 Literature

The model of the effectiveness of monetary policy in the presence of a fiscal regime that controls spending was introduced by Sargent & Wallace (1981). Even under the standard monetarist assumption of long-run money neutrality (i.e., the assumption that an increase in the money supply leads, in the long run, to higher inflation), the ability of the monetary authority to control the path of inflation is not guaranteed. If the government accumulates debt financed through persistent fiscal deficits, the central bank may eventually be forced to accommodate these liabilities through money creation, thereby losing effective control over the rate of money growth. Without the ability to control money growth, the monetary authority is unable to control inflation: the resulting path will be determined jointly by central bank money creation and the monetization of government debt.

This contribution laid the foundations for Leeper (1991), who developed a simple model: an endowment economy with monetary and fiscal policy. The monetary authority reacts to inflation by raising interest rates proportionally to deviations of inflation from target, while the fiscal authority collects taxes (there is an assumption of no government spending in this paper), which are also proportional to deviations of debt from its desired target. Policy is described as "active" when it is not constrained by these stabilizing rules. For the monetary authority, active policy means raising interest rates by more than the increase in inflation. For the fiscal authority, active policy means abandoning the surplus rule that forces debt to remain at its target level. As a result, four possible regimes arise: (i) active monetary policy and active fiscal policy, (ii) active monetary policy and passive fiscal policy, (iii) passive monetary policy and active fiscal policy, and (iv) passive monetary policy and passive fiscal policy. Regimes (i) and (iv) do not deliver a determinate equilibrium. By contrast, regime (ii), commonly referred to as the monetary-led (ML) regime, and regime (iii), known as the fiscally-led (FL) regime, are associated with unique equilibria. Leeper derived the theoretical properties of these equilibria and established the conditions under which the two regimes can be distinguished.

Because the different authorities determine the paths of inflation and debt in each regime, it is important to determine in which regime the country currently operates. This importance is particularly highlighted in Bianchi & Melosi (2019), since policies that are optimal in one regime may be detrimental in another. In their example, the central bank may respond to higher inflation by raising interest rates. However, if the government does not adjust primary surpluses to stabilize debt, the result is a policy conflict: higher debt, lower output, and higher inflation than under either a clearly monetary-led or fiscally-led regime. This provides the primary motivation for research on fiscal and monetary dominance, as it helps explain the joint dynamics of inflation and output.

Research on the existence and probability of fiscal and monetary regimes has generally followed two distinct approaches. The first, employed by Sims (2010), involves estimating a

Markov-Switching Vector Autoregression (MS-VAR) model, with particular attention paid to the relationships between deficits and debt, as well as between interest rates and inflation. This methodology was subsequently applied by Kliem (2016) to the United States, Germany, and Italy, and by Bianchi & Melosi (2017) to the United States. These studies were able to identify clear differences in central bank responses to inflation and in government debt dynamics. The classical approach to estimating MS-VAR models, used by the authors above, relies on Bayesian methods with non-informative priors. Sims & Zha (2006) provide an important methodological contribution using U.S. data: they introduce a measure of fiscal stance (the primary deficit-to-debt ratio), which is also used in this paper.

However, as both Kliem (2016) and Bianchi & Melosi (2017) note, the MS-VAR framework may serve only as initial evidence of regime changes. This is because the regime is ultimately determined by a set of deeper structural parameters derived from a general equilibrium model. To this end, it is necessary to develop a Dynamic Stochastic General Equilibrium framework. The methods used to estimate MS-DSGE models are also well established in the literature. For standard DSGE models, Herbst & Schorfheide (2016) provide a comprehensive introduction to the construction of Dynamic Stochastic General Equilibrium models, Bayesian inference, and Markov Chain Monte Carlo techniques, with particular emphasis on the Metropolis–Hastings algorithm for DSGE estimation. These methods are employed in the MS-DSGE framework developed in this paper.

The current mainstream approach in DSGE modeling relies on rational expectations (RE), both in forward-looking and hybrid specifications. Classical rational expectations assume that agents can identify the equilibrium path of the economy and respond accordingly. In this context, agents must take into account the inherent regime switching governed by a Markov process. To address this issue, Maih (2015) introduced Rationality in Switching Environments (RISE), a toolbox that allows rational expectations to be formed in models with regime-dependent parameters. The key idea is that agents recognize that regimes evolve according to a Markov process and therefore incorporate the probability of regime transitions into their expectations and decision-making.

The literature on fiscal and monetary rules for Poland, including regime-switching behavior, also provides an important foundation for the calibration and interpretation of the model developed in this paper. Brzoza-Brzezina, Kolasa & Szetela (2016) estimated a DSGE model for Poland, and several of their calibrated parameters are used in the present study. Their work focused primarily on the possibility of Poland reaching the zero lower bound, an event that would likely correspond to a period of fiscal dominance. However, they concluded that such a scenario was relatively unlikely for Poland, given the country's comparatively strong growth performance and a higher inflation target than in many Western European economies, which reduced the risk of persistent deflation. Their analysis, however, did not incorporate time-varying coefficients or regime switching behavior. The issue of switching regimes was also examined by Baranowski & Sztaudynger (2016), who developed a Markov-switching Taylor rule for Poland. Their analysis did not directly address fiscal dominance, but they found evidence that the central bank changed its responsiveness to inflation and the output gap over time. Nevertheless, the differences between regimes were relatively moderate, and the authors did not identify an underlying transition mechanism beyond the estimated evolution of policy coefficients. Regarding DSGE estimations for Poland, Grabek, Kłos, & Koloch (2011) provide a useful reference both for calibrating certain parameters and for constructing prior distributions for Poland, which I employ in this paper.

3 Motivating evidence

Before creating full model, I want to see whether there was a regime-switching behavior visible in the data. To this end, I develop VAR(1) model, in which I allow the coefficients to change. I use four observables to estimate the Markov-Switching Vector Autoregression (MS-VAR): (i) inflation; (ii) real GDP growth rate; (iii) real overnight interest rate ex ante; (iv) ratio of primary surplus to current debt.

I allow for Markov-switching regimes for the parameters of VAR matrix and for the variance-covariance matrix (Σ_t). I estimate the model looking for two regimes Φ_{ξ_t}: each with its own Φ_t and associated Σ_t. My main purpose is distinguishing between the parameters (especially the persistence of surplus/debt [for fiscal block] and relationship between interest rate and inflation [for monetary block]) in order to find the probabilities of different regimes in each time. The estimated model can be characterized by the set of equations:

$$Z_t = c_{\xi_t} + A_{\xi_t} Z_{t-1} + \Sigma_{\xi_t}^{\frac{1}{2}} \varepsilon_t \tag{1}$$

$$\Phi_{\xi_t} = (c_{\xi_t}, A_{\xi_t}, \Sigma_{\xi_t}), \varepsilon_t \sim N(0, I) \tag{2}$$

which is in fact a VAR(1).

To estimate the VAR model for Poland, I develop a Bayesian Vector Autoregression (BVAR) framework. In a standard VAR, each variable is explained by its own past values and the past values of all other variables in the system. In a Bayesian VAR, instead of treating the coefficients as fixed unknown parameters, they are treated as random variables with prior probability distributions. These priors are then updated using the data, resulting in posterior distributions that combine prior beliefs with empirical information.

To begin the estimation, I use the Minnesota prior, which is one of the most commonly used priors in BVAR models. The intuition behind the Minnesota prior is simple: it assumes that each variable is primarily driven by its own recent past, and that more distant lags or other variables play a smaller role. Concretely, the prior typically "pulls" each variable toward a random walk, meaning that own first lags are expected to be important, while cross-variable effects are expected to be smaller unless the data strongly suggests otherwise.

In a Bayesian VAR (BVAR) with regime switching (such as an MS-VAR), the smoothed posterior probability of a regime at a given date is the probability—computed using the full sample of data—that the economy was in that particular regime at that time, conditional on all available information in the sample (both past and future observations). Unlike filtered probabilities, which only use information up to time t, smoothed probabilities incorporate the entire dataset, which typically makes them more stable and informative about the underlying latent state. The resulting smoothed posterior probabilities of the regimes in MS-VAR are visible in the 1. The lighter variation corresponds to the regime 1, while the darker to the regime 2.

Figure 1: Posterior probability of policy regimes in MS-VAR

Table 1: Estimated Regime-Specific VAR Coefficients

Coefficient	Regime 1	Regime 2
Real rate response to inflation $\phi_{r,\pi}$	0.9521	0.1012
Surplus/deficit persistence $\phi_{f,f}$	0.9903	0.1801
Surplus/deficit constant $\phi_{f,0}$	0.0226	-0.3881

As for the regimes found: the results of the estimates for the most important parameters are given in the table 1. Regime 1 has a very strong response of real rate to inflation. It is also connected with positive $\phi_{f,0}$ (in the steady state, the government raises the surplus), as well as higher persistence of this variable.

I find further arguments for this distinction looking at macroeconomic variables in both regimes. Regime 1 is characterized by moderate inflation averaging 2.3%, strong real GDP growth of 3.8%, and positive real interest rates of 1.9%. In contrast, Regime 2 exhibits substantially higher inflation of 6.7%, weaker growth of 2.1%, and strongly negative real interest rates of -3.0%. Fiscal conditions also deteriorate in Regime 2, as indicated by a more negative surplus/debt ratio.

Table 2: Average Macroeconomic Conditions by Estimated Policy Regime

Variable	Regime 1	Regime 2
Inflation (%)	2.30	6.70
Real GDP Growth (%)	3.83	2.07
Real Interest Rate (%)	1.91	-3.00
Surplus/debt (%)	-3.96	-5.97

Taking into account that regime 1 corresponds to stronger reaction to both debt (which means stronger fiscal adjustments), as well as stronger reaction of real rates to inflation, while regime 2 shows neither a strong fiscal consolidation nor strong real rate reaction to inflation, it is therefore probable that regime 1 corresponds to the monetary led (ML) regime, while regime 2 is connected with fiscally led (FL) regime.

Additionally, it is useful to include the resulting posterior probability of regime change matrix, presented in 6. Those values have later been used to create a prior for MS-DSGE.

Table 3: Posterior Transition Matrix for MS-VAR

$$P = \begin{bmatrix} 0.9881 & 0.0119 \\ 0.1299 & 0.8701 \end{bmatrix}$$

Considering all this information, MS-VAR gives us a strong reason to look into the period 2019-2023, where the strongest and most probable period of regime 2 took place. However, because of lack of additional constraints it is not possible to determine the character of these two regimes. Even with the full information, VAR model cannot be used to find the structural parameters. In order to check my findings regarding the presence and character of regimes, I create a general equilibrium model, that includes the fiscal-monetary block.

4 Model

In this section I first present a simplified model of endowment economy with fiscal and monetary authority, following Leeper (1991), in which I derive 2 possible unique conditions for each regime. In this economy parameters are constant - there is no regime switching. It serves to explain the nation of regimes and their importance. I then develop New Keynesian model, which serves as the main model in this paper.

4.1 A stochastic endowment economy

A representative infinitely-lived consumer receives a constant endowment of Y units of the non-storable consumption good each period. They derive utility U from consumption C_t of it, with discount factor b is affected by an exogenous process ε_t^d that follows an independent and identically distributed (i.i.d.) zero–mean exogenous process. The government issues one term bonds to households at purchase price Q_t, which can trade it for one unit of goods at price P_t, and collects lump-sum taxes τ_t. As such, the target of the household is to choose a path of (C_t, B_t) that maximizes lifelong utility:

$$\max_{\{C_t,B_t\}} E_0 \sum_{t=0}^{\infty} \beta^t \text{exp}(\varepsilon_t^d) U(C_t) \tag{3}$$

subject to the flow budget constraint $P_tC_t + Q_tB_t + P_t\tau_t = P_tY_t + B_{t-1}$ with $Q_t < 1$. Solving this problem yields an Euler equation, which simplifies to Fisher equation in this economy:

$$R_t = E_t(\beta^{-1} \frac{\text{exp}(\varepsilon_t^d)}{\text{exp}(\varepsilon_{t+1}^d)} \Pi_{t+1}) \tag{4}$$

The monetary policy sets nominal interest rate $R_t = Q_t^{-1}$ by using the rule:

$$\frac{R_t}{R^*} = (\frac{\Pi_t}{\Pi^*})^{\psi} \tag{5}$$

The government budget constraint holds:

$$B_tQ_t + P_t\tau_t = B_{t-1} \tag{6}$$

which can be also rewritten as:

$$\tau_t + R_t^{-1} b_t = \frac{b_{t-1}}{\Pi_t} \tag{7}$$

,
where $b_t = B_t/P_t$ denotes the real value of government debt.

The government is willing to raise taxes (or primary surplus - in this economy they are equivalent) according to the fiscal rule:

$$\tau_t - \tau^* = \delta(b_{t-1} - b^*) + \varepsilon_t^\tau \tag{8}$$

where $E(\varepsilon_t^\tau) = 0$.

I linearize the model around the steady state (hatted variables are deviation from their steady state levels, while tilded variables are log-deviations) and express the Fisher and Taylor equations:

$$\tilde{R}_t = \varepsilon_t^d + E_t\tilde{\pi}_{t+1} \tag{9}$$

$$\tilde{R}_t = \psi\tilde{\pi}_t \tag{10}$$

The same way I present the fiscal bloc

$$\hat{\tau}_t + \hat{b}_t = \beta^{-1}\hat{b}_{t-1} + b^*\tilde{R}_t - \beta^{-1}b^*\tilde{\pi}_t \tag{11}$$

$$\hat{\tau}_t = \delta\hat{b}_{t-1} + \varepsilon_t^\tau \tag{12}$$

The equation (11) serves as a link between fiscal and monetary policy. Inflation, which path is dependent on the ψ parameter set by monetary authority enters into fiscal bloc, thus creating a fiscal–monetary interaction.
Combining equations (9) and (10) into one yields

$$\psi\tilde{\pi}_t = \varepsilon_t^d + E_t\tilde{\pi}_{t+1} \tag{13}$$

Doing the same with equations (10), (11) and (12) gives

$$\hat{b}_t = (\beta^{-1} - \delta)\hat{b}_{t-1} + (\psi - \beta^{-1})b^*\tilde{\pi}_t - \varepsilon_t^\tau \tag{14}$$

I present it in one matrix equation:

$$\begin{bmatrix} E_t(\tilde{\pi}_{t+1}) \\ b_t \end{bmatrix} = \begin{bmatrix} \psi & 0 \\ \psi - \beta^{-1})b^* & \beta^{-1} - \delta \end{bmatrix} \begin{bmatrix} \tilde{\pi}_t \\ b_{t-1} \end{bmatrix} + \begin{bmatrix} -\varepsilon_t^d & 0 \\ 0 & -\varepsilon_t^\tau \end{bmatrix} \tag{15}$$

This is the final fiscal–monetary bloc of this model. It has 2 eigenvalues - ψ and $\beta^{-1} - \delta$. There is one state variable (determined in the previous period) and one jump variable (which is able to change in this period following shocks). Following Blanchard & Kahn (1980), the existence of unique path requires therefore one eigenvalue in the unit root and one outside it. This exercise confirms the results of parameter space partition introduced by Leeper (1991). When $\psi > 1$ and $\beta^{-1} - \delta < 1$, we are in a monetary–led regime. The fiscal authority is keeping the debt in check, while the monetary authority keeps inflation down. The opposite ($\psi < 1$ and $\beta^{-1} - \delta > 1$) denotes the fiscally–led regime. In this case, the government does not react to growing debt sufficiently, while the central bank does not raise rates in reaction to inflation. The price level is determined by the interaction between real debt and real surpluses.

Because the regimes are defined only by those parameters, learning about the regime in the data means estimating their values. This is precisely the task of the next section.

4.2 New Keynesian model

Following this simplified example, I develop Markov Switching Dynamic Stochastic General Equilibrium model, which is augmented typical New Keynesian (NK) model with added fiscal block and the Markov process that generates two regimes of monetary-fiscal policy. This NK model, based on the model used by Bianchi & Melosi (2017), involves representative agents: households, firms and government, which has 2 branches: monetary that controls the (nominal) interest rate and fiscal which controls spending.

4.2.1 Households

A representative household maximizes expected lifetime utility:

$$\max_{\{C_t, N_t, B_t\}} E_0 \sum_{t=0}^{\infty} \beta^t \text{exp}(\zeta_t) \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} \right) \tag{16}$$

subject to the nominal budget constraint:

$$P_t C_t + B_t \leq W_t N_t + R_t B_{t-1} + \Omega_t - T_t \tag{17}$$

where W_t is the nominal wage, N_t is labour supplied to the market, Ω_t is the aggregate profit, σ is inverse elasticity of intertemporal substitution, φ is the inverse Frisch elasticity of labor supply and $\zeta_t = \rho_\zeta \zeta_{t-1} + \varepsilon_t^\zeta$ is a preference shock.

4.2.2 Firms

Assume a continuum of firms indexed by $i \in [0, 1]$. Each firm produces a differentiated good, but they all use an identical technology, represented by the production function:

$$Y_t(i) = A_t N_t(i) \tag{18}$$

The technology A_t grows according to the rule:

$$A_t = (1 + \Gamma)^t \tag{19}$$

where Γ denotes the long-term growth.

A competitive final-good firm bundles a continuum of differentiated intermediate goods into a final composite:

$$Y_t = \left(\int_0^1 Y_t(i)^{\frac{\varepsilon-1}{\varepsilon}} \, di \right)^{\frac{\varepsilon}{\varepsilon-1}} \tag{20}$$

,

with demand for each intermediate goods given as:

$$Y_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon} Y_t \tag{21}$$

Price setting follows the Calvo (1983) mechanism. Each period, a fraction $(1-\theta)$ of firms receives permission to re-optimise their price; the remaining fraction keep their current price unchanged. A firm choosing P_t^* solves:

$$\max_{P_t^*} E_t \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} \left[P_t^* Y_{t+k|t} - P_{t+k} MC_{t+k} Y_{t+k|t} \right] \tag{22}$$

subject to the demand function for variety i given above, where $Q_{t,t+k}$ is stochastic discount factor between t and $t+k$, while MC_{t+k} denotes the nominal marginal cost.

4.2.3 Fiscal-monetary block

The monetary authority reacts to inflation, as well as an output gap. Both of those reactions are dependent on the regime in place - in the monetary regime the reaction to both is larger than in the fiscal regime, because the monetary authority is passive then. Markov process ξ_t determines the current regime in place.

The Taylor rule can therefore be written as:

$$R_t = R^* \left(\frac{R_{t-1}}{R^*} \right)^{\rho_r} \left[\left(\frac{\Pi_t}{\Pi^*} \right)^{\phi_\pi^{\xi_t}} \left(\frac{Y_t}{Y^*} \right)^{\phi_y^{\xi_t}} \right]^{1-\rho_r} \exp\left(\sigma_r \varepsilon_t^{MP}\right) \tag{23}$$

.

In turn fiscal authority collects taxes every period, issues new debt, repays the debt outstanding and spends the money as a government consumption.

$$B_t = R_{t-1} B_{t-1} + P_t G_t - P_t T_t \tag{24}$$

which may be presented in real terms and in relation to output ($b_t = \frac{B_t}{P_t Y_t}$)

$$b_t = \frac{R_{t-1} Y_{t-1}}{\Pi_t Y_t} b_{t-1} + g_t - t_t \tag{25}$$

I assume that taxation is kept constant in this period, and the change in fiscal policy (and resulting surpluses $s_t = t_t - g_t$ is driven entirely by government expenditure. As such, the fiscal rule is as follows:

$$g_t - g^* = \rho_g \left(g_{t-1} - g^*\right) - \delta_b(1-\rho_g)\left(b_{t-1} - b^*\right) - \delta_y(1-\rho_g)\log\left(\frac{Y_t}{Y^*}\right) + \sigma_g \varepsilon_t^g \tag{26}$$

4.2.4 Equilibrium

A competitive equilibrium is a set of processes $\{C_t, N_t, Y_t, Y_t(i), P_t(i), P_t, W_t, R_t, B_t, G_t, T_t\}_{t=0}^{\infty}$ such that households and firms optimize, price-setting is optimal, and all markets clear.

Household optimality The household optimality conditions include:

$$C_t^{-\sigma} = \beta\, \mathbb{E}_t\left[C_{t+1}^{-\sigma}\frac{R_t}{\Pi_{t+1}}\right] \tag{27}$$

$$\frac{W_t}{P_t} = \frac{N_t^{\varphi}}{C_t^{-\sigma}} \tag{28}$$

Goods market clearing Final output is used for consumption and government spending:

$$Y_t = C_t + G_t \tag{29}$$

Price index and aggregation

$$Y_t = \left(\int_0^1 Y_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di\right)^{\frac{\varepsilon}{\varepsilon-1}} \tag{30}$$

$$Y_t(i) = \left(\frac{P_t(i)}{P_t}\right)^{-\varepsilon} Y_t \tag{31}$$

Fiscal block Government debt evolves according to:

$$b_t = \frac{R_{t-1}}{\Pi_t \Gamma_y} b_{t-1} + g_t - t_t \tag{32}$$

and fiscal policy follows the rule:

$$g_t - g^* = \rho_g(g_{t-1} - g^*) - \delta_b(1-\rho_g)(b_{t-1} - b^*) - \delta_y(1-\rho_g)\log\left(\frac{Y_t}{Y^*}\right) + \sigma_g \varepsilon_t^g \tag{33}$$

Monetary policy The nominal interest rate follows:

$$R_t = R^* \left(\frac{R_{t-1}}{R^*}\right)^{\rho_r} \left[\left(\frac{\Pi_t}{\Pi^*}\right)^{\phi_\pi^{\xi_t}} \left(\frac{Y_t}{Y^*}\right)^{\phi_y^{\xi_t}}\right]^{1-\rho_r} \exp\left(\sigma_r \varepsilon_t^{MP}\right) \tag{34}$$

Market clearing Bond market clears:

$$B_t = B_t^H \tag{35}$$

Labor market clears:

$$N_t = \int_0^1 N_t(i)\, di \tag{36}$$

Goods market:

$$Y_t = \int_0^1 Y_t(i)\, di = C_t + G_t \tag{37}$$

4.2.5 Loglinear model

Now I can describe variables as deviations (described by hat) or log-deviations (described as tilde) from the steady state. Additionally, the consumption and product are log-linearized around the steady-state trend, not the cycle $Y_t = (1+\Gamma)^t Y^*(1+\tilde{y}_t)$.

The resulting New Keynesian model may be altogether given in a set of 7 equations:

- New Keynesian IS curve

$$\tilde{y}_t = E_t(\tilde{y}_{t+1}) - \frac{1}{\sigma}(\hat{i}_t - E_t(\hat{\pi}_{t+1})) + \gamma_g'(\hat{g}_t - E_t\hat{g}_{t+1}) + \zeta_t - \sigma_y^{COVID}\varepsilon_t^{COVID} \tag{38}$$

- New Keynesian Philips curve

$$\hat{\pi}_t = \beta(1+\Gamma)^{\frac{1}{\sigma}} E_t\hat{\pi}_{t+1} + \kappa\tilde{y}_t + u_t + +\sigma_\pi^{COVID}\varepsilon_t^{COVID} \tag{39}$$

- Monetary rule

$$\hat{i}_t = \rho_i \hat{i}_{t+1} + (1-\rho_i)(\phi_\pi^{\xi_t}\hat{\pi}_t + \phi_y^{\xi_t}\tilde{y}_t) + \varepsilon_t^{MP} \tag{40}$$

- Fiscal rule

$$\hat{g}_t = \rho_g \hat{g}_{t-1} - \delta_b^{\xi_t}\hat{b}_t - \delta_g \tilde{y}_t + \varepsilon_t^g \tag{41}$$

- Evolution of government debt

$$\hat{b}_t = (1+i^* - \pi^* - \Gamma)\hat{b}_{t-1} + (i^* + \hat{i}_{t-1} - \pi^* - \hat{\pi}_t - \Gamma - \tilde{y}_t + \tilde{y}_{t-1})b^* + g^* + \hat{g}_t + \varepsilon_t^b \tag{42}$$

- Evolution of demand shock

$$\zeta_t = \rho_\zeta \zeta_{t-1} + \varepsilon_t^\zeta \tag{43}$$

- Evolution of cost-push shock

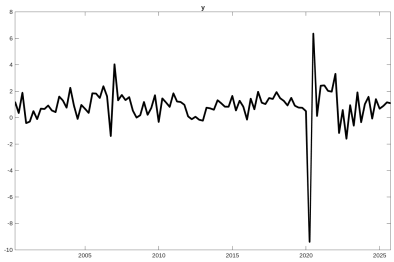
$$u_t = \rho_u u_{t-1} + \varepsilon_t^u \tag{44}$$

I added two shocks motivated by evidence: the Covid-19 shock, denoted by ε_t^{COVID} (which was set from Q2-2020 to Q2-2021) with negative impact on product and positive impact on inflation. The size of this shock was estimated in the model. This way, proposed by Smets (2026) was used in order to reduce the effects of Covid-19 pandemic on the estimation. Additionally, the debt shock ε_t^b was also used to calibrate the size of government debt, in order to account for the structural breaks, of which I speak in the data section, as well as imperfect data on government deficit and effective rates.

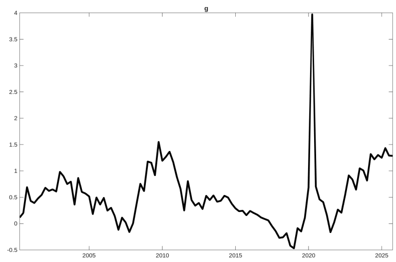
Given this model, each regime can be defined. In the regime 1 the $\phi_\pi^{\xi_t=1} > 1$ and $\delta_b^{\xi_t=1} > \beta^{-1}(1+\Gamma)^{-\frac{1}{\sigma}} - 1$, while regime 2 is the opposite: $\phi_\pi^{\xi_t=2} < 1$ and $\delta_b^{\xi_t=2} < \beta^{-1}(1+\Gamma)^{-\frac{1}{\sigma}} - 1$. This time, unlike in MS-VAR, the regimes may be correctly identified thanks to the structural parameters of the model.

5 Data

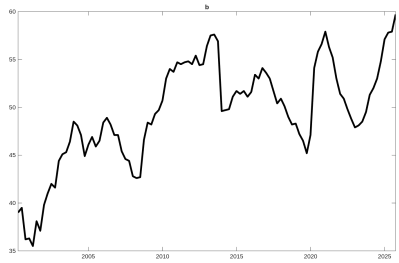
The empirical analysis is based on five quarterly macroeconomic time series: the inflation rate, the real GDP growth rate, the short-term nominal interest rate, the ratio of government debt to GDP, and the ratio of the primary surplus to GDP. The sample spans the period from 2000-Q2 to 2025-Q4, yielding a total of 103 quarterly observations. The starting point of the sample is determined by the construction of the inflation series, which requires the previous quarter's price level and therefore cannot be computed for 2000-Q1. All the variables are presented below.



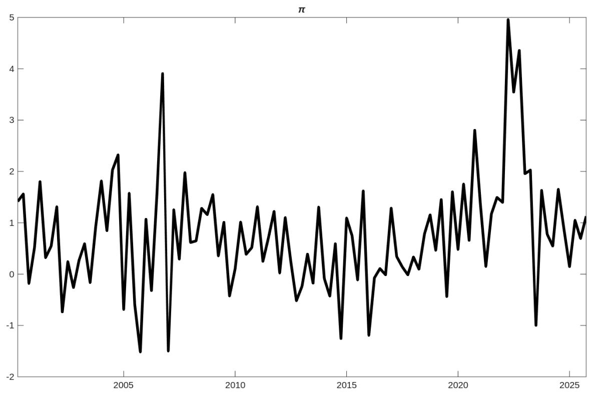
(a) Real GDP growth



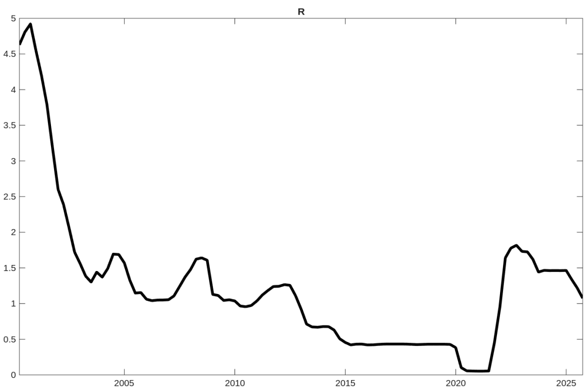
(b) Primary deficit



(c) Government debt



(d) Inflation



(e) Interest rates

Figure 2: The chosen variables from 2000Q2 to 2025Q4, in percentage points.

The short-term nominal interest rate is represented by the three-month interbank rate for Poland (FRED series: IR3TIB01PLQ156N). Government indebtedness is measured using quarterly general government debt data obtained from Eurostat (Eurostat series: gov_10q_ggdebt). It should be noted that this series is affected by several structural breaks resulting from methodological and institutional changes, particularly major fiscal reforms. The most important example is the 2013 pension reform, under which a substantial share of OFE pension fund assets and liabilities was transferred outside the general government sector, leading to a discrete decline in the reported debt level.

Inflation and real output growth are constructed as log differences of the GDP deflator (FRED series: NGDPDSAIXPLQ) and real GDP (FRED series: NGDPRSAXDCPLQ), respectively. Since the model assumes the existence of a non-zero steady-state growth path, the cyclical component of real GDP growth is extracted using the Hodrick–Prescott filter in order to isolate deviations from trend growth.

The primary surplus series is constructed from general government net lending/borrowing data (FRED series: GGNLBAPLA188N). Because this measure includes government interest payments, net interest expenditures are removed in order to obtain the primary balance. Net interest payments are calculated as the difference between government interest expenditures and interest revenues using Eurostat data (Eurostat series: gov_10q_ggnfa). The resulting series is subsequently seasonally adjusted to eliminate the pronounced seasonal component present in fiscal variables.

All observable variables are demeaned or transformed by removing a steady state (this is the case for inflation, for which I assume the quarterly steady state at 0.625%). The variables used, along with their description, are described in the table 4.

Table 4: Variables used in the analysis

Variable	**Description**	**Source**	**Mean**
Inflation	Log difference of GDP deflator	IMF	0.78%
Real GDP growth	HP-filtered log difference of real GDP	IMF	0.9%
Short-term interest rate	90-day interbank rate	OECD	1.24%
Debt-to-GDP ratio	Quarterly general government debt	Eurostat	49.42%
Primary surplus-to-GDP ratio	General government net lending/borrowing adjusted for net interest payments	FRED / Eurostat	-0.53%

6 Results

6.1 Calibration and Bayesian Estimation

I calibrate some parameters (namely: risk aversion (σ), household discount factor (β), Philips curve slope (κ)) from literature, set some from the policy rules (steady-state inflation (π^*), response of government expenditure to debt in fiscal regime ($\phi_{\pi}^{\xi_t=1}$) and average quarterly GDP growth (Γ)) from the previous estimations, while at the same time estimating the rest of them from the available data.

The entire model has been created in RISE framework, with the set of 7 equations, 2 regimes and 24 variables, as well as 7 endogenous variables (5 of which are observable) and 5 exogenous variables (one of which is observable). Chosen parameters has been calibrated using the results of Grabek, Klos, Koloch (2011), while the rest has been given prior distribution. The full set of used priors and calibrated parameters is given in appendix. Following this process, the Markov Chain Monte Carlo (specifically Metropolis-Hastings algorithm) with N=1M draws was done in order to get posterior distribution of the parameters and estimate the placing of the shocks.

Firstly, the model is able to reproduce the movements of the Polish economy in the given period. As I show in the next picture, the estimated model is in line with the data.

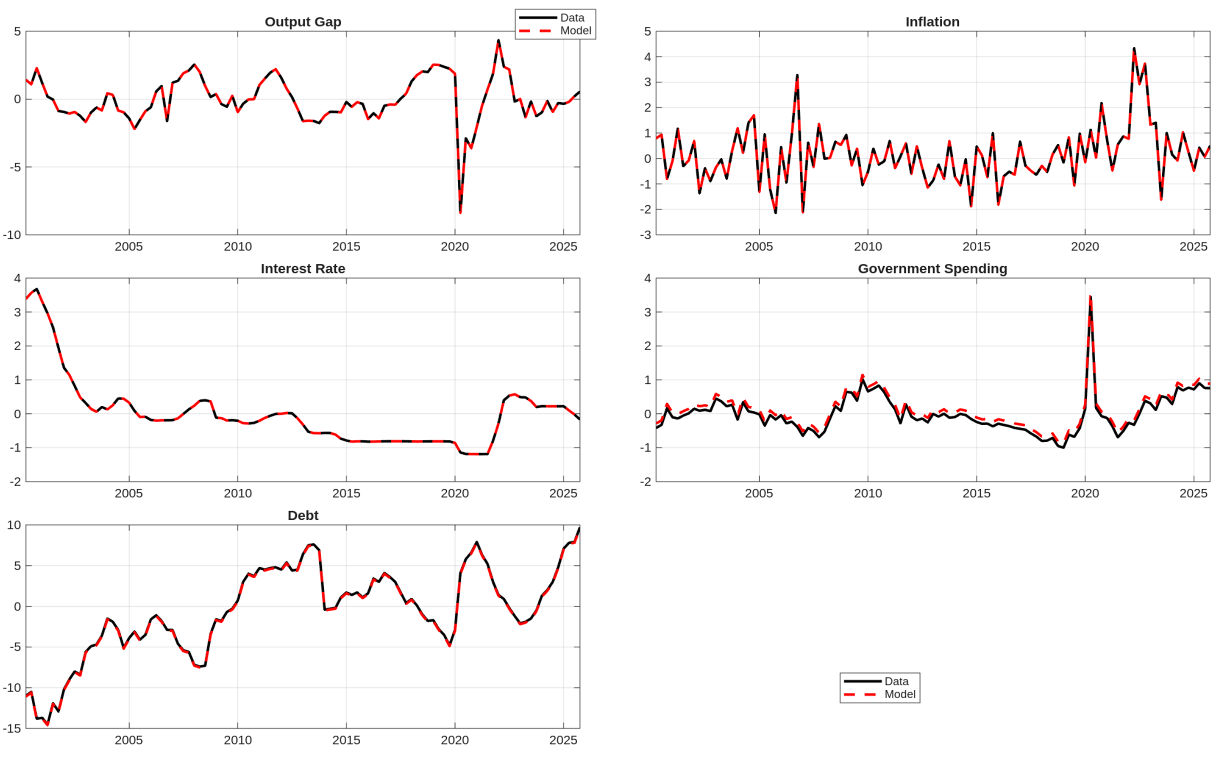


Figure 3: The estimated model compared to the data. On the y axis is the deviation from the mean in percentage points.

The theoretically correct acceptance ratio in MCMC is 0.234. In the estimation I used this parameter was about 0.26, which means that the Bayesian estimate was close to theoretically sound. The analysis of prior and posterior distribution also shows significant differences, which mean that the analysis was influenced by data, not only by the choice of priors, which I show in tables 10 and 11.

The resulting regimes are clearly different. This is especially visible when comparing their defining characteristic: reaction of interest rates to deviation inflation and of government expenditures to debt. Those parameters are shown in table 5.

Table 5: Estimated Regime-Specific DSGE Parameters

Coefficient	Regime 1	Regime 2
Interest rate response to inflation ϕ_π	1.5761	0.7130
Government spending response to debt δ_b	0.0480	0

The resulting smoothed probabilities of the regimes in MS-DSGE following burn-in of 10% of posterior results are visible in the figure 4.

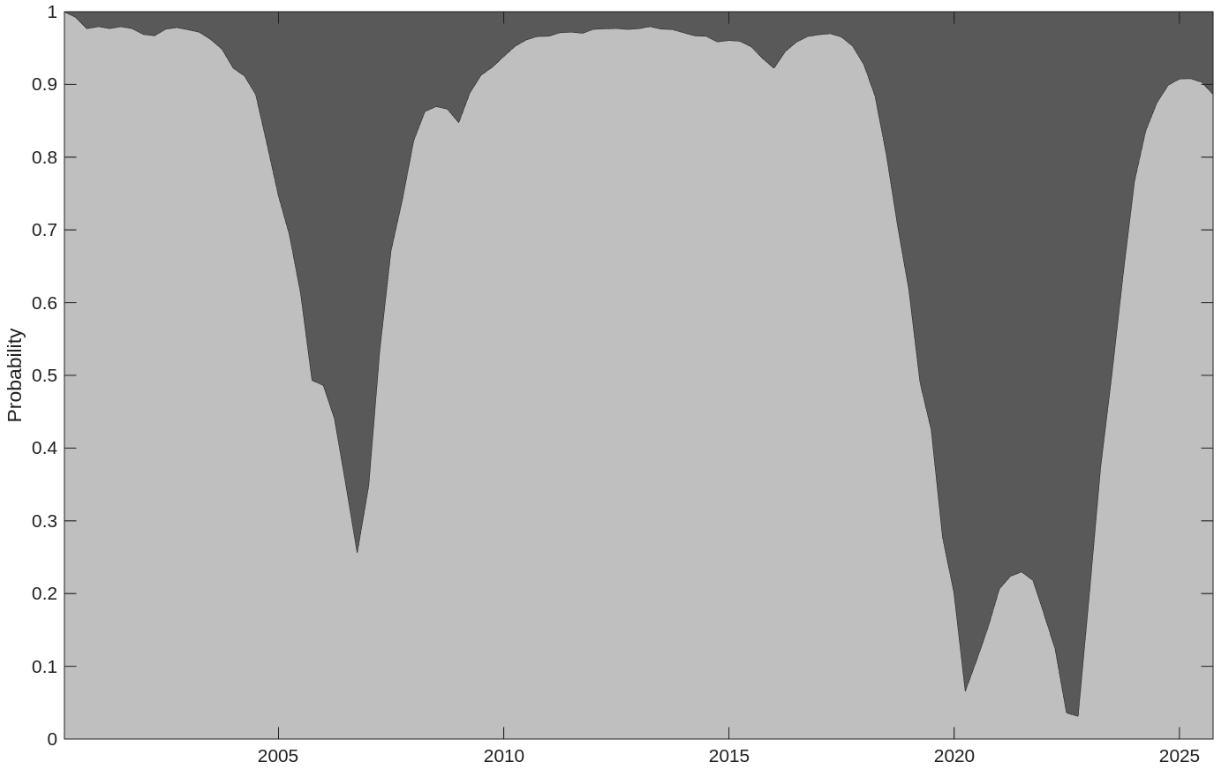


Figure 4: Posterior probability of policy regimes in MS-DSGE

Although the chance of either regime being in place is never 0, the posterior probability of regime 2 (which is a fiscal regime) exceeds 50% twice. First time is around the year 2006

- following the entry into European Union, the inflation rose, and the central bank deemed it transitory (Baranowski, Sztaudynger 2016). There is also a second period - from 2019Q1 to 2023Q2, and reaches its maximum in 2022Q2. This is also to be expected - as Bianchi & Melosi (2019) have shown, the reaction to negative demand shock is often the switch to fiscal dominance.

Markov process which rules the regime change is not symmetrical. As a result, the time during which regime is in place is also different for each regime. The estimated posterior transition matrix can be seen here:

Table 6: Posterior Transition Matrix for MS-DSGE

$$P = \begin{bmatrix} 0.9695 & 0.0305 \\ 0.1292 & 0.8708 \end{bmatrix}$$

As we can see, the dominant regime is regime 1 (monetary-led) - it is more stable and lasts longer than fiscally led regime, which happens (on average) rarer and lasts shorter.

6.2 Behavior of the model

The standard way of illustrating regime-dependent transmission mechanisms is through impulse response functions (IRFs), which trace the dynamic response of endogenous variables to a one-standard-deviation structural shock. In this paper, I focus on four shocks: a monetary policy shock (ε_t^{MP}), a fiscal policy shock (ε_t^{g}), a demand shock (ε_t^{ζ}), and a cost-push shock (ε_t^{u}). The full set of IRFs is presented in the Appendix: the figure 7 for monetary regime and 8 for fiscal regime.

In the monetary regime the primary stabilizing force in response to shocks operates through the nominal interest rate. While fiscal policy also reacts, it is ultimately the real interest rate channel that drives the economy back toward steady state, with fiscal policy largely focused on stabilizing government debt dynamics. In particular, when inflationary pressures arise, the central bank responds by raising the nominal interest rate more aggressively, thereby containing inflation and stabilizing the economy.

In the second regime fiscal policy plays the dominant role in determining the price level. Inflation adjusts endogenously to ensure that government debt remains bounded over time. Importantly, fiscal policy does not actively stabilize the debt level through primary surpluses; instead, the adjustment burden is shifted to inflation dynamics. As a consequence, the nominal interest rate increases by less than inflation, implying a decline in the real interest rate.

To show this difference, we can see the reaction of inflation to the same demand shock in figure 5.

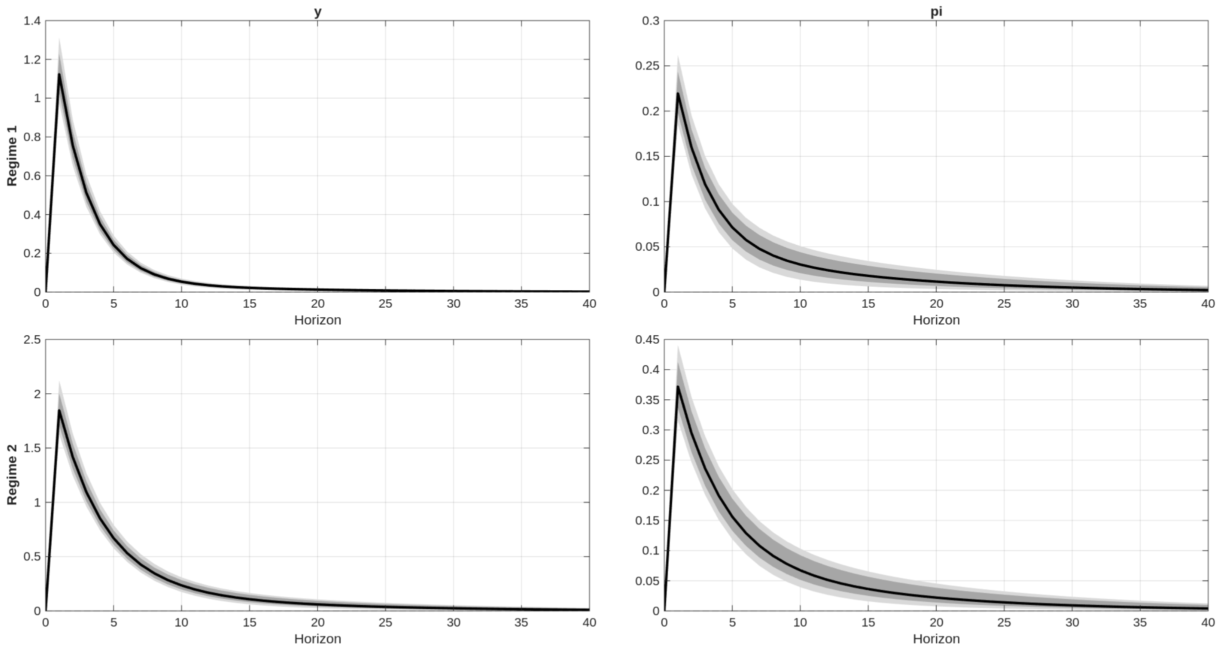


Figure 5: Impulse Response Function of output gap and inflation gap (in percentage points) to 1s.d. demand shock. On the top is the monetary regime, on the bottom is the fiscal regime.

Because the real rate is growing in the regime 1 (for the monetary authority reacts to inflation by sharply raising interest rates) and falling in the regime 2, the reaction to shock is almost 2 times larger in the fiscal regime.

I show similar IRFs in figure 6, this time for cost-push shock.

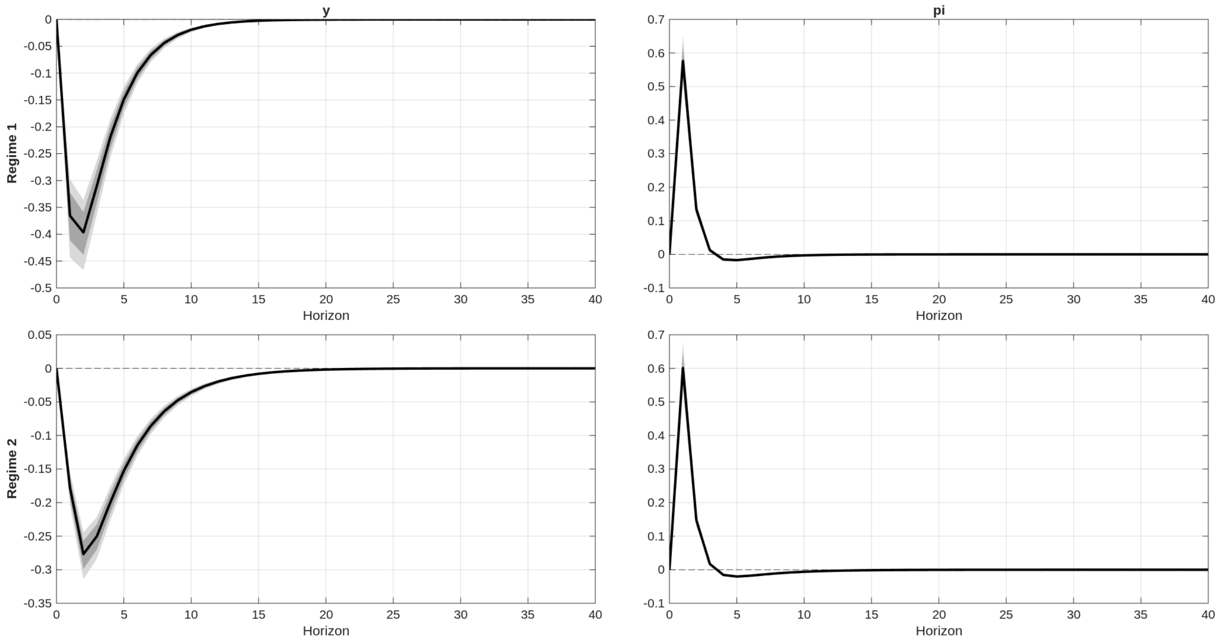


Figure 6: Impulse Response Function of output gap and inflation gap (in percentage points) to 1s.d. cost-push shock. On the top is the monetary regime, on the bottom is the fiscal regime.

Here the drop in production on impact is larger in the case of monetary regime, while inflation behaves similarly in both regimes. However, the monetary regime goes back to the steady-state significantly faster.

The created IRFs are consistent with the literature and theory of the model - the reaction of interest rates, inflation and movement of debt across the regimes is in line with the literature regarding fiscal and monetary regimes. This serves as a robustness check for the estimation.

It is noteworthy that the estimated posterior value of ϕ_π in the fiscal regime remains relatively high (around 0.8). This reflects the institutional reality that Poland has never operated under a formal fiscal dominance arrangement (as was the case for the United States prior to 1951). As a result, even during periods of elevated inflation, the central bank continues to respond, albeit insufficiently to fully stabilize inflation dynamics.

7 Conclusions

Both MS-VAR and MS-DSGE show a clear distinction between two possible regimes: the first was the standard in this period, while the second happened around the year 2006 as well as from 2019 to 2022. According to the model the differing regimes were, respectively, monetary led and fiscally led. During fiscally led periods, the inflation generally speaking rose, the central bank did not adjust the rates accordingly, while the general government allowed the high level of debt to endure.

The fiscal dominance is not always the same. Leeper (2023) himself distinguished 3 different parts of fiscally led regimes: unwitting dominance (in which the government want to keep the debt down, even when central banks try to activate economy, as following the “fiscal compact” from 2012 in the EU), insidious dominance (when the debt limit is changing, and yet government is not adjusting correctly, as is the case with rapidly aging countries) and political dominance (when the central bank is forced to act as a banker for the government to keep the bonds low, despite the rules saying otherwise). This analysis does not provide differentiation between those types of fiscal dominance periods in Poland.

It must also be noted, that this analysis is not centered on the welfare. Without it, it is not possible to decide whether regime change - which amplified inflation, forced the interest rate to remain at a higher level for longer and increased the government debt - was net positive or net negative. As Leeper (2023) himself notes: "There is nothing inherently “bad” about fiscal dominance or “good” about monetary dominance". It is very well possible, that switch from monetary led regime into fiscally led regime following Covid-19 pandemic or change in acceptable debt level following Great Recession has been a correct way to deal with threat of hitting zero lower bound (ZLB), as Bianchi & Melosi (2017) predict for the United States.

The main conclusion from this paper is that according to the model Poland entered into a period of fiscal dominance even before the Covid-19 pandemic and associated economic crisis, and was there throughout most of the crisis thereafter. The central bank did not respond strongly to inflation, while government disregarded the level of debt. This period ended following the strong monetary policy response to inflation in the course of 2022. However, this did not result in the fiscal consolidation, which may mean that the fiscal and monetary policies are currently in conflict, as is often the case following the crisis. As Bianchi & Melosi (2019) proved in their paper - the conflict between fiscal and monetary authority results in neither achieving their goal. Continued imbalances and increased lack of certainty regarding the ruling regime would be detrimental to macroeconomic stability.

References

[1] Baranowski, P., & Sztaudynger, J. (2016). Polish monetary policy through the lens of regime switching Taylor rule. *Ekonometria* 3(53).

[2] Bianchi, F., & Melosi, L. (2017). Escaping the Great Recession. *American Economic Review*, 107(4), 1030–1058.

[3] Bianchi, F., & Melosi, L. (2019). The Dire Effects of the Lack of Monetary and Fiscal Coordination. *Journal of Monetary Economics*, 104, 1–22.

[4] Blanchard, O., & Kahn, C. (1980). The Solution of Linear Difference Models under Rational Expectations. *Econometrica*, 48(5), 1305–1312

[5] Brzoza-Brzezina, M., Kolasa, M., & Szetela, P. (2016). Is Poland at risk of zero lower bound? *Bank i Kredyt 47(3)*, 2016, 195–226.

[6] Calvo, G. A. (1983). Staggered Prices in a Utility-Maximizing Framework. *Journal of Monetary Economics*, 12(3), 383–398.

[7] Galí, J. (2008). *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework*. Princeton University Press.

[8] Grabek, G., Kłos, B., & Kołoch, G. (2011). SOEPL 2009 – An Estimated Dynamic Stochastic General Equilibrium Model for Policy Analysis And Forecasting *NBP Working Papers* 83, Narodowy Bank Polski.

[9] Herbst, E., & Schorfheide, F. (2016). *Bayesian Estimation of DSGE Models*. Princeton University Press.

[10] Kliem, M., Kriwoluzky, A., & Sarferaz, S. (2016). On the Low-Frequency Relationship Between Public Deficits and Inflation. *Journal of Applied Econometrics*, 31(3), 566–583.

[11] Leeper, E. M. (1991). Equilibria Under "Active" and "Passive" Monetary and Fiscal Policies. *Journal of Monetary Economics*, 27(1), 129–147.

[12] Leeper, E. M. (2023). *Monetary, Fiscal, and Debt Management Policy Interactions*. NBER Working Paper.

[13] Maih, J. (2015). Efficient Perturbation Methods for Solving Regime-Switching DSGE Models. *Norges Bank Working Paper*.

[14] Powell, J. H. (2025). Speech at the Jackson Hole Economic Symposium. Federal Reserve Bank of Kansas City.

[15] Reis, R. (2026). Why did inflation rise and fall in 2021-24? Channels and evidence from expectations. *LSE Working Paper*.

[16] Sargent, T. J., & Wallace, N. (1981). Some Unpleasant Monetarist Arithmetic. *Federal Reserve Bank of Minneapolis Quarterly Review*, 5(3), 1–17.

[17] Sims, C. A. (1994). A Simple Model for Study of the Determination of the Price Level and the Interaction of Monetary and Fiscal Policy. *Economic Theory*, 4(3), 381–399.

[18] Webb, C. (2025). Fiscal Dominance in the Modern Era: Revisiting Sargent and Wallace in Light of UK Macroeconomic History *LSE Working Paper*.

[19] Woodford, M. (1994). Monetary Policy and Price Level Determinacy in a Cash-in-Advance Economy. *Economic Theory*, 4(3), 345–380.

A VAR coefficients of MS-VAR

Table 7: Estimated Coefficient Matrix — Regime 1 (Monetary Dominance)

$$A_1 = \begin{bmatrix} -0.0008 & 0.8793 & 0.0254 & 0.0036 & -0.0002 \\ 0.0036 & -0.0130 & 0.7385 & -0.0135 & -0.0019 \\ 0.1976 & 0.9510 & 0.0013 & 1.0211 & 0.0141 \\ 0.0211 & -0.0032 & 0.0028 & -0.0023 & 0.9896 \end{bmatrix}$$

Table 8: Estimated Coefficient Matrix — Regime 2 (Fiscal Dominance)

$$A_2 = \begin{bmatrix} -0.0008 & 0.8793 & 0.0254 & 0.0036 & -0.0002 \\ 0.0036 & -0.0130 & 0.7385 & -0.0135 & -0.0019 \\ -0.2334 & 0.1005 & -0.0036 & 0.9794 & 0.0013 \\ -0.4379 & 0.0030 & -0.0038 & -0.0045 & 0.1668 \end{bmatrix}$$

B Prior distribution of MS-DSGE

Table 9: Calibrated Parameters

Parameter	Description	Value
σ	Inverse intertemporal elasticity of substitution	1.00
β	Discount factor	0.99
κ	Slope of the Phillips curve	0.05
$\delta_b^{(2)}$	Fiscal feedback in regime 2	0.00
π_{ss}	Steady-state quarterly inflation	0.63%
Γ	Deterministic quarterly growth rate	0.9%

Table 10: Prior Distributions for Estimated Parameters

Parameter	Prior Mean	Initial Value	Std. Dev.	Distribution	Lower Bound	Upper Bound
ρ_r	0.75	0.75	0.05	Beta	0.30	0.98
ρ_u	0.50	0.50	0.10	Beta	0.00	0.95
ρ_g	0.60	0.60	0.10	Beta	0.00	0.98
ρ_ζ	0.80	0.80	0.08	Beta	0.30	0.99
b_{ss}	0.55	0.55	0.03	Normal	–	–
R_{ss}	0.010	0.010	0.002	Normal	–	–
g_{ss}	0.020	0.020	0.003	Normal	–	–
γ	0.08	0.08	0.03	Normal	0.00	0.50
δ_y	0.05	0.05	0.03	Normal	0.00	0.50
$\phi_\pi^{(1)}$	1.80	1.80	0.15	Normal	1.10	3.00
$\phi_\pi^{(2)}$	0.85	0.85	0.08	Normal	0.50	1.00
$\phi_y^{(1)}$	0.25	0.25	0.08	Normal	0.00	1.00
$\phi_y^{(2)}$	0.10	0.10	0.05	Normal	0.00	0.30
$\delta_b^{(1)}$	0.08	0.08	0.03	Gamma	0.01	0.50
σ_r	0.01	0.01	0.005	Inverse Gamma	–	–
σ_ζ	0.01	0.01	0.005	Inverse Gamma	–	–
σ_u	0.01	0.01	0.005	Inverse Gamma	–	–
σ_g	0.01	0.01	0.005	Inverse Gamma	–	–
σ_b	0.01	0.01	0.005	Inverse Gamma	–	–
$\sigma_{covid,y}$	0.03	0.03	0.01	Inverse Gamma	–	–
$\sigma_{covid,\pi}$	0.02	0.02	0.01	Inverse Gamma	–	–
p_{12}	0.02	0.02	0.01	Beta	–	–
p_{21}	0.12	0.12	0.03	Beta	–	–

C Posterior distribution of MS-DSGE

I present the full posterior mean of estimated DSGE parameters in table 11.

Table 11: Posterior Parameter Estimates

Parameter	Mean	Median	LB68	UB68	LB90	UB90
ρ_r	0.8821	0.8832	0.8666	0.8976	0.8553	0.9063
ρ_u	0.3009	0.3001	0.2556	0.3471	0.2251	0.3749
ρ_g	0.5539	0.5543	0.4966	0.6128	0.4506	0.6483
ρ_ζ	0.9167	0.9183	0.9000	0.9337	0.8862	0.9428
b_{ss}	0.3032	0.3025	0.2626	0.3435	0.2403	0.3719
R_{ss}	0.0097	0.0097	0.0077	0.0117	0.0063	0.0131
g_{ss}	0.0034	0.0034	0.0021	0.0047	0.0013	0.0056
γ	0.0765	0.0765	0.0474	0.1052	0.0301	0.1241
δ_y	0.0840	0.0828	0.0583	0.1098	0.0451	0.1273
$\phi_{\pi,1}$	1.5761	1.5745	1.4061	1.7470	1.2975	1.8600
$\phi_{\pi,2}$	0.7130	0.7121	0.6312	0.7952	0.5838	0.8427
$\phi_{y,1}$	0.4099	0.4109	0.3598	0.4613	0.3226	0.4923
$\phi_{y,2}$	0.1440	0.1471	0.0892	0.1930	0.0562	0.2318
$\delta_{b,1}$	0.0480	0.0458	0.0319	0.0637	0.0248	0.0781
σ_r	0.0021	0.0021	0.0019	0.0023	0.0018	0.0024
σ_ζ	0.0030	0.0029	0.0025	0.0035	0.0023	0.0039
σ_u	0.0046	0.0046	0.0043	0.0049	0.0041	0.0052
σ_g	0.0083	0.0083	0.0077	0.0090	0.0073	0.0095
σ_b	0.0140	0.0140	0.0131	0.0151	0.0125	0.0158
$\sigma_{covid,y}$	0.0408	0.0394	0.0275	0.0546	0.0222	0.0645
$\sigma_{covid,\pi}$	0.0172	0.0165	0.0118	0.0226	0.0095	0.0272
$snk_{tp,1,2}$	0.0323	0.0305	0.0201	0.0445	0.0147	0.0566
$snk_{tp,2,1}$	0.1309	0.1292	0.1014	0.1605	0.0864	0.1815

D Impulse Response Functions

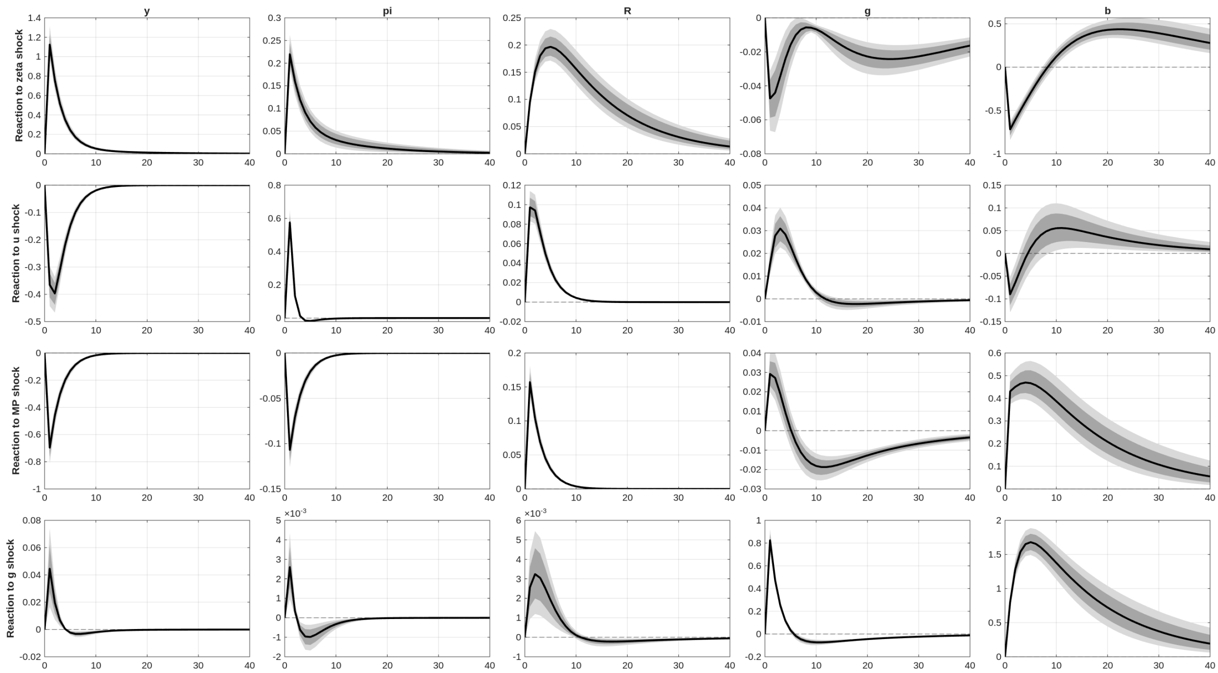


Figure 7: Impulse response functions to one-standard-deviation shocks: demand, cost-push, monetary policy, and government spending shocks in the monetary-led regime.

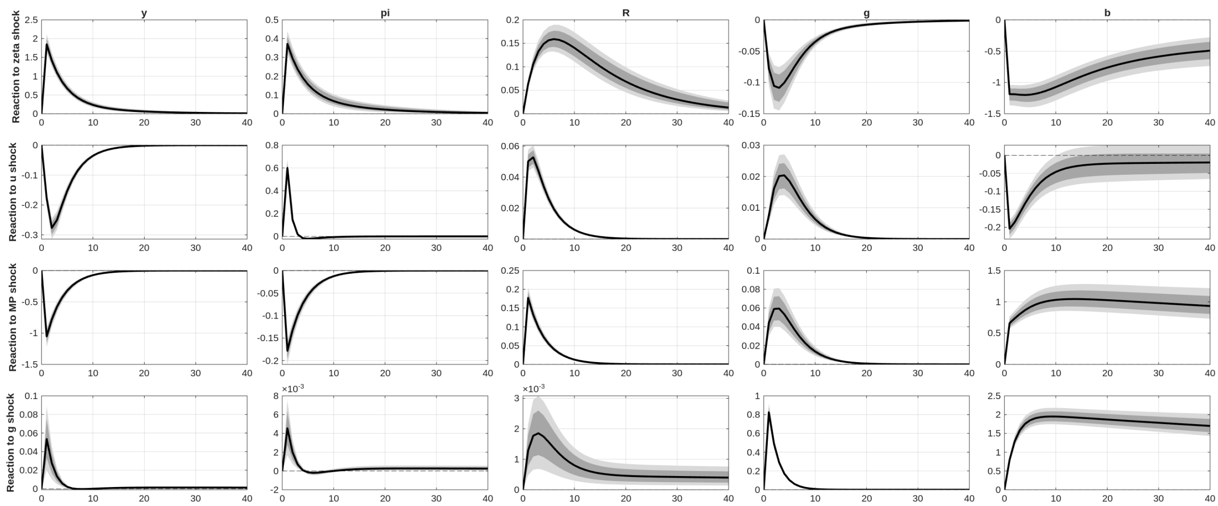


Figure 8: Impulse response functions to one-standard-deviation shocks: demand, cost-push, monetary policy, and government spending shocks in the fiscally-led regime.